

1. Marked relation complex

$$\begin{aligned} P_1 &= \mathbb{Q} \xrightarrow{d_1(z)=(0,0,z)} P_0 = \mathbb{Q}^3 \\ G_1 &= 3, \quad G_0 = \text{diag}(1, 2, 3), \quad G_0 d_1 = d_1 G_1 \end{aligned}$$



2. Homology, effective algebra and seed closure

$$\begin{aligned} H_0(P_\bullet) &= \mathbb{Q}^2 = F, \quad \iota = I, \quad \bar{G} = \text{diag}(1, 2) \\ \mathcal{A}_{\text{eq}}^{\text{eff}} &= \mathbb{Q}[\bar{G}] = \text{the two-dimensional diagonal algebra} \\ R_0(s) &= (s, 0), \quad N = N_{\text{eq}} = \mathbb{Q}(1, 0) \end{aligned}$$



3. Supplied state and source maps

$$\begin{aligned} M = R &= \mathbb{Q}^3, \quad T = A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{pmatrix}, \quad \Lambda = I \\ D(x_1, x_2, x_3) &= (0, x_3), \quad q(y_1, y_2) = y_2 \\ qD\Lambda &= H = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \end{aligned}$$



4. Stable assembly and the exact floor

$$\begin{aligned} \ker \Lambda &= 0, \quad W = U = \mathbb{Q}e_2, \quad T|_U = 2I_U \\ \chi_{\text{forced}} &= \mu_{\text{forced}} = S - 2 \\ \text{Landing descent holds; normalized closure fails because } Ae_1 &= e_3. \end{aligned}$$

Here q identifies F/N with $\mathbb{Q}$. This is one explicit marked realization. It does not canonically supply the inputs for every inverse-Leibniz presentation.